In the right angled $\Delta AMC$, $\tan 30° = \dfrac{CM}{AM}$

$$\frac{1}{\sqrt{3}} = \frac{CM}{140}$$

$$CM = \frac{140}{\sqrt{3}} = \frac{140\sqrt{3}}{3}$$

$$= \frac{140 \times 1.732}{3}$$

$$CM = 80.83$$

Now, $h = CD = CM + MD = 80.83 + 60 = 140.83$

Therefore, the height of the second building is $140.83$ m

**Example 6.28** From the top of a tower $50$ m high, the angles of depression of the top and bottom of a tree are observed to be $30°$ and $45°$ respectively. Find the height of the tree. $(\sqrt{3} = 1.732)$

***Solution*** The height of the tower $AB = 50$ m

Let the height of the tree $CD = y$ and $BD = x$

From the diagram, $\angle XAC = 30° = \angle ACM$ and $\angle XAD = 45° = \angle ADB$

In the right angled $\Delta ABD$,

$$\tan 45° = \frac{AB}{BD}$$

$$1 = \frac{50}{x} \Rightarrow x = 50 \text{ m}$$

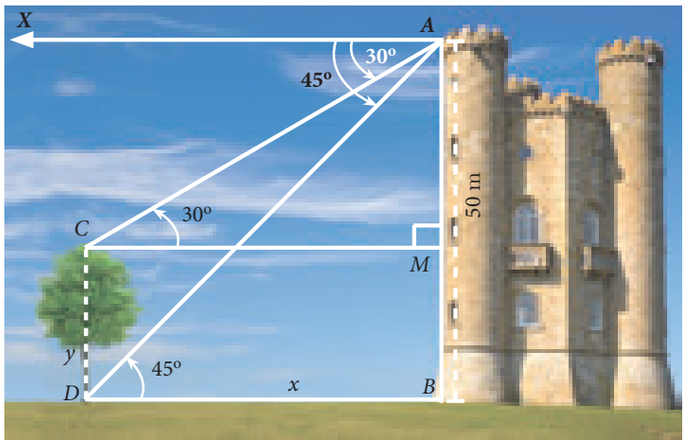



Fig. 6.23

In the right angled $\Delta AMC$,

$$\tan 30° = \frac{AM}{CM}$$

$$\frac{1}{\sqrt{3}} = \frac{AM}{50} \quad [\because DB = CM]$$

$$AM = \frac{50}{\sqrt{3}} = \frac{50\sqrt{3}}{3} = \frac{50 \times 1.732}{3} = 28.87 \text{ m.}$$

Therefore, height of the tree = $CD = MB = AB - AM$ = 50 – 28.87 = 21.13 m



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